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Dynamic vehicle routing problem with drone resupply for same-day delivery☆

Juan C. Pina-Pardo [a,b], Daniel F. Silva [c,*], Alice E. Smith [c], Ricardo A. Gatica [a]

[a] School of Industrial Engineering, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile
[b] Center for Transportation and Logistics, Massachusetts Institute of Technology, Cambridge, USA
[c] Department of Industrial and Systems Engineering, Auburn University, Auburn, USA

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ABSTRACT

We study a same-day delivery problem where customer orders arrive dynamically throughout the day and the service operator must determine, in real-time, whether to accept the orders and how to adjust the ongoing distribution plan. We develop a route-based Markov Decision Process and an efficient online policy to dynamically route a truck that can receive newly arrived orders along its route via drones dispatched from a depot. Numerical experiments show that our online policy has an average fill rate decrease of at most 20% over the perfect-information counterpart. Further, this online policy has a fill rate increase of up to 8% over a naïve greedy policy. We also show that drone resupply increases fill rates by up to 21% compared to a conventional truck-only resupply system. Computational times to make each decision are in the hundredths of a second, thus allowing real-time feedback to customers regarding their eligibility for same-day delivery.

1. Introduction

Same-Day Delivery (SDD) enables customers to place and receive orders on the same day. The convenience of shopping online without going through the typical inconveniences associated with the traditional brick-and-mortar experience has made SDD a game-changer in the world of e-commerce in recent years. As a result, SDD is one of the fastest-growing segments in the last-mile delivery context, with annual growth rates of nearly 30% (World Economic Forum, 2020).

Since the last mile is the most expensive stage of the e-commerce logistics chain (Business Insider, 2018), many companies are considering and investing in new technologies, such as drones, to reduce their logistics costs while meeting the increased demand for speedy delivery (Moshref-Javadi and Winkenbach, 2021). There are multiple factors that make drones attractive: their low capital expenditure costs per vehicle, their reduced carbon footprint, their ability to perform autonomous deliveries, and their fast point-to-point travel capabilities (Gaba and Winkenbach, 2022).

In this paper, we investigate the use of drones to resupply delivery vehicles with newly arrived orders, enabling SDD providers to inject these orders into ongoing distribution routes without requiring delivery trucks to return to the depot to collect them. To this end, we formally introduce the *Dynamic Traveling Salesman Problem with Drone Resupply* (DTSP-DR), which consists of dynamically routing a single delivery truck that can receive newly arrived orders along its route via drones dispatched from a depot. In the DTSP-DR, a set of orders is available at the depot at the beginning of the day (e.g., next-day delivery orders from the previous day

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* Corresponding author.
E-mail address: silva@auburn.edu (D.F. Silva).

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or overnight orders), while others dynamically arrive throughout the delivery horizon. The objective of the DTSP-DR is to maximize the expected number of orders delivered within the operational horizon.

Due to its dynamic nature, we formulate the DTSP-DR as a MDP (Powell, 2019; Ulmer et al., 2020). The proposed MDP formulation provides a comprehensive description of the problem, but it also follows that the DTSP-DR suffers from the "curse of dimensionality" in both the state and action spaces. Therefore, we also devise an online policy based on reoptimization, which provides decisions regarding incoming orders in real-time. This paper is the first to propose a dynamic mathematical model for drone resupply with orders arriving dynamically and real-time decision-making for SDD requests in terms of acceptance/rejection and route updates. Drone resupply avoids the issues involved with drones interacting directly with customers while providing the benefits of drones in the final supply chain leg – improved operational efficiency and flexibility at a low investment cost, along with better sustainability. Our solution methodology can be readily applied in practice for real-time applications such as e-commerce, where customers are presented with delivery options (including same-day service) during the checkout process.

The remainder of this paper is organized as follows. We review the extant literature on SDD and highlight our contributions in Section 2. Section 3 presents the MDP formulation. Section 4 describes the static counterpart of the DTSP-DR and presents an upper bound calculation. Our solution approach is described in Section 5. We then present the test instances and numerical experiments in Section 6. Finally, we report our conclusions in Section 7.

2. Literature review

The DTSP-DR belongs to the class of Dynamic Vehicle Routing Problems (DVRPs) with applications in SDD. While there is increasing attention in the academic literature on tactical and operational SDD problems (see, e.g., Zhang and Van Woensel (2022)), here we summarize the works most related to the DTSP-DR. Klapp et al. (2018) studied an SDD problem where a single truck must deliver orders that arrive dynamically throughout the day. The authors formulated the problem as an MDP, where decisions are made at discrete moments of the day (referred to as dispatch waves). The authors also developed a rollout policy and discussed the trade-off between minimizing total costs and maximizing the total number of orders delivered. In Ulmer et al. (2019), the authors investigated the benefits of allowing a single delivery truck to perform preemptive return trips to the depot to pick up new orders. They formulated the problem as an MDP and proposed an approximate value iteration procedure combined with the Cheapest Insertion technique (Rosenkrantz et al., 1977) to update the ongoing routing plan. In both the MDP and the solution approach, the authors make decisions every time the truck stops at a customer location, which delays order acceptance decisions until such an event occurs.

More recently, Voccia et al. (2019) addressed an SDD problem with multiple trucks and delivery time windows. The authors defined the problem as an MDP, where decisions associated with each new order consist of waiting to assign the order (which assumes that order acceptance decisions can be postponed) or assigning it either to one of the delivery trucks or to a third-party logistics provider. They also presented a scenario-based heuristic to incorporate information about future orders into dispatching decisions. Archetti et al. (2020) studied the problem of finding a collection of delivery routes for a single truck to deliver orders whose release dates (i.e., the time when orders arrive at the depot from upstream distribution facilities) are stochastic. Similar to Klapp et al. (2018), the authors did not allow preemptive depot returns, so dispatching decisions are not made in real-time but only when the truck is at the depot (i.e., at the beginning of the day and every time the truck returns to the depot). Archetti et al. (2020) developed a reoptimization and a myopic solution approach to analyze the trade-off between solution quality and the computational effort required.

Combining trucks and drones for last-mile logistics has been widely studied in static routing settings (see, e.g., Macrina et al. (2020) and Moshref-Javadi and Winkenbach (2021) for recent surveys). One of the first works to investigate using drones to reduce last-mile delivery costs was presented by Murray and Chu (2015), who studied a routing problem where a truck and drone cooperatively deliver customer orders. Since then, most of the existing contributions have focused on extending the work of Murray and Chu (2015) mainly by considering multiple trucks and drones (Murray and Raj, 2020; Sacramento et al., 2019; Ham, 2018; Kitjacharoenchai et al., 2019), by allowing the drones to perform multiple deliveries per sortie (Gonzalez-R et al., 2020; Luo et al., 2021; Schermer et al., 2019; Schaumann et al., 2024), and by considering drone stations (Chauhan et al., 2019; Zhu et al., 2022). Note that this literature assumes a drone can deliver directly to the customer, a situation that we believe is fraught with difficulties such as safety and human behavior (Cornell et al., 2023; Gaba and Winkenbach, 2020).

To the best of our knowledge, Ulmer and Thomas (2018) were the first to study the use of drones for SDD, where orders arrive dynamically and stochastically during the day. In the proposed delivery system, both trucks and drones serve customers, but drones do so directly on round trips from the depot (i.e., drones do not work in tandem with ground delivery trucks). They modeled the problem as an MDP and developed a Policy Function Approximation (PFA) to determine whether a truck or a drone will serve a given customer. Their numerical results discussed how incorporating drones could reduce the number of trucks needed during the day. Extending Ulmer and Thomas (2018) and Chen et al. (2022) developed a deep Q-learning solution approach to learn the value of taking an action at each system state. Using the assignment and routing heuristics from Ulmer and Thomas (2018) to update the ongoing distribution plan, Chen et al. (2022) showed that their deep Q-learning approach outperforms the PFA proposed by Ulmer and Thomas (2018).

Our research is closely related to the works presented by Dayarian et al. (2020), Pina-Pardo et al. (2021), and Pina-Pardo et al. (2024). To the best of our knowledge, Dayarian et al. (2020) were the first to propose using drones for resupplying newly arrived SDD orders. The authors studied a delivery system composed of a single delivery truck and a single resupply drone, where the objective is to maximize the total number of customers served during the operational horizon. Without proposing a mathematical formulation (e.g., an MDP formulation) to model the underlying dynamic routing decisions, Dayarian et al. (2020) showed the benefits of drone resupply using only heuristic approaches, where decisions are made at previously defined decision epochs (thus delaying order acceptance decisions until such moments occur). Pina-Pardo et al. (2021) studied a static routing problem where all customer information (i.e., delivery locations and order arrival times) is known before the operation starts. Considering a single delivery truck and a single resupply drone, the authors showed that drone resupply can reduce the total duration of the delivery process (i.e., the makespan) by up to 20%. Finally, Pina-Pardo et al. (2024) extended the static problem studied by Pina-Pardo et al. (2021) to consider multiple trucks and resupply drones, where newly released orders can be collected by trucks via depot returns or resupplied by drones en route. The authors developed a Mixed-Integer Linear Programming (MILP) formulation and proposed a unified matheuristic approach that can solve both the truck-and-drone and truck-only scenarios. Pina-Pardo et al. (2024) showed the effectiveness of the matheuristic approach and derived managerial insights for instances with up to 100 customers.

One of the most challenging and realistic aspects of our work is considering real-time decision-making, as customers expect immediate feedback on their eligibility for SDD (Klapp et al., 2020). To the best of our knowledge, only five works in the extant DVRP literature consider real-time decision-making (i.e., Azi et al. (2012), Ichoua et al. (2006), Ulmer (2020), Ulmer and Thomas (2020), and Zhang et al. (2023)). Among these papers, only Azi et al. (2012) and Zhang et al. (2023) reported the computational times to make each decision, and both of these exceeded several seconds. None of these works considered drone resupply.

2.1. Research gaps and contributions

Our literature review reveals that most SDD works make decisions at discrete moments of the day, which is clearly operationally impractical for real-time applications. Moreover, the vast majority of the existing literature on drone logistics has focused on using drones for direct delivery to customers (Macrina et al., 2020), even though the protocols and regulations on how customers would actually receive orders from drones do not yet exist for most last-mile applications (Cornell et al., 2023). In response to these research gaps, we make the following contributions.

First, we formally introduce and mathematically formulate the DTSP-DR, the first dynamic model developed for a truck routing problem with drone resupply. Second, we formulate a MILP model for the static counterpart of the DTSP-DR to derive an upper bound on the expected number of orders delivered during the day. Third, we propose an efficient online policy, based on reoptimization, which embeds a fast MILP model to adjust drone resupply operations optimally. Fourth, we show that this online policy has an average fill rate decrease of at most 20% over the perfect-information counterpart and that computational times to make each decision are in the hundredths of a second. We also show that this online policy has a fill rate increase of up to 8% over a greedy policy where any order that can be feasibly served is accepted. Finally, we show that drone resupply increases fill rates by up to 21% compared to a conventional truck-only resupply system.

3. Markov decision process formulation

In the DTSP-DR, we consider a single depot, a single uncapacitated delivery truck, a fleet of capacitated resupply drones, a set of orders available at the depot at the beginning of the day (e.g., next-day delivery orders held over from the previous day or overnight orders), and a set of orders that arrive dynamically throughout the day. Each dynamic order is characterized by an arrival time and a delivery destination (in the following, we use the terms customer, order, and destination interchangeably). Following standard practices of SDD providers, we consider that the dispatcher must immediately decide whether to accept or reject a customer's request for same-day delivery of an order. Applications of this system can typically be found in e-commerce (e.g., Amazon Fresh), where customers are presented with delivery options (including SDD based on an ongoing distribution plan) during the checkout process, after entering their delivery address (Klapp et al., 2020). Thus, accepted orders are those for which same-day service is offered (and therefore must be delivered within the delivery horizon), and rejected orders are those for which same-day service is not offered (these customers are offered delivery on a later day). Alternative assumptions are that rejected orders are lost (Ulmer et al., 2019; Zhang et al., 2023) or served by an external third-party logistics company (Voccia et al., 2019). In case of acceptance, the dispatcher must determine how to insert the newly arrived order into the current truck route, as well as the timing and location where a drone will resupply the order to the truck. There is a fixed time imposed for each drone unloading at the truck, and meeting locations of the truck and a drone are restricted to customer locations along the route. All newly arriving orders that are accepted for SDD are resupplied to the truck by drones throughout the day (as in Dayarian et al. (2020) and Pina-Pardo et al. (2021)). The objective of the DTSP-DR is to maximize the expected number of orders delivered within an operational horizon, consistent with research studies on SDD (see, e.g., Ulmer and Thomas (2018) and Voccia et al. (2019)). Note that, since the DTSP-DR aims to maximize the number of orders delivered, not offering same-day service may be interpreted as a penalty paid by the dispatching organization.

We formulate the DTSP-DR as an MDP following the route-based modeling paradigm of Ulmer et al. (2020). Note that the problem definition does not assume that orders arrive according to a specific stochastic process, or that the parameters of the arrival process are known. However, formulating the DTSP-DR as an MDP is still possible because the model proceeds in discrete time with the decision epochs corresponding to the arrival times of new orders. Since all other aspects of the problem are deterministic, the Markov property is preserved for the MDP even if the raw arrival process is not memoryless (see, e.g., Ghiani et al. (2022)).

The components of the MDP formulation are described below (all notation introduced throughout this section is summarized in Table A.11 of Appendix).

3.1. Decision epochs and system states

We consider that each decision epoch τ occurs every time a new order arrives. By convention, the first decision epoch occurs at the beginning of the day and the second when the first dynamic order arrives, and so on.

The system state s_τ at decision epoch τ summarizes all the information available to the dispatcher. We define $s_\tau := (\lambda^\tau, \mathcal{A}^\tau, \mathrm{T}^\tau, \mathrm{D}^\tau)$, where λ^τ is the time at which decision epoch τ occurs, $\mathcal{A}^\tau$ is the set of accepted orders that are available at the depot, T^τ is a vector that captures the truck information, and D^τ is a vector that captures the drone information. Table 1 details the components of the truck and drone information vectors.

Table 1
Components of the truck and drone information vectors.

Symbol	Description
Truck vector information	
T^τ	Truck vector information at decision epoch τ. Formally, $\mathrm{T}^\tau := (\mathcal{X}^\tau, \mathcal{L}^\tau)$.
$\mathcal{X}^\tau$	Truck route visiting nodes in $\mathcal{N}^\tau$ at decision epoch τ .
$\mathcal{N}^\tau$	Set of nodes visited in the current truck route. Formally, $\mathcal{N}^\tau := \mathcal{L}^\tau \cup \mathcal{O}^\tau \cup \mathcal{A}^\tau \cup \{0\}$.
$\mathcal{L}^\tau$	Set of orders currently loaded onto the truck at decision epoch τ.
$\mathcal{O}^\tau$	Set of orders currently onboard all drones at decision epoch τ.
Drone vector information	
K	Set of resupply drones.
D^τ	Drone vector information at decision epoch τ. Formally, $\mathrm{D}^\tau := (\mathrm{D}_k^\tau)_{k\in K}$.
D_k^τ	Vector information of drone $k \in K$ at decision epoch τ. Formally, $\mathrm{D}_k^\tau := (\alpha_k^\tau, \mathcal{O}_k^\tau, \mathrm{P}_k^\tau)$.
α_k^τ	Time when drone $k \in K$ will be available at the depot to start a new sortie.
$\mathcal{O}_k^\tau$	Set of orders onboard drone $k \in K$ at decision epoch τ.
P_k^τ	Vector of planned sorties for drone $k \in K$ at decision epoch τ. Formally, $\mathrm{P}_k^\tau := (\mathrm{P}_{k(i)}^\tau)_{i\in \mathcal{I}_k^\tau}$.
$\mathcal{I}_k^\tau$	Set of sortie indices associated with drone $k \in K$ at decision epoch τ.
$\mathrm{P}_{k(i)}^\tau$	The ith planned sortie. Formally, $\mathrm{P}_{k(i)}^\tau := (\mathcal{R}_{k(i)}^\tau, m_{k(i)}^\tau, \xi_{k(i)}^\tau)$.
$\mathcal{R}_{k(i)}^\tau$	Subset of orders to be resupplied to the truck in the ith sortie of drone $k \in K$.
$m_{k(i)}^\tau$	Meeting location (with the truck) associated with the ith sortie of drone $k \in K$.
$\xi_{k(i)}^\tau$	Scheduled launch time of the ith sortie of drone $k \in K$.

Note that each accepted (but not yet delivered) order belongs to exactly one of these sets: $\mathcal{L}^\tau$, $\mathcal{O}^\tau$, or $\mathcal{A}^\tau$, meaning that orders are either on the truck, on a drone, or at the depot, respectively. Furthermore, note that planned drone schedules will not necessarily be executed before the next decision epoch (i.e., planned drone sorties can be modified in future decision epochs). Finally, note that the current truck route considers visiting all accepted orders, regardless of whether the orders are currently loaded onto the truck.

3.2. Actions and rewards

When a new order arrives (denoted by ϑ^τ), an action a_τ must be selected based on the system state s_τ. Formally, we define $a_\tau := \left(z^\tau, \mathrm{T}^{\tau,a}, \mathrm{D}^{\tau,a}\right)$, where z^τ is an order acceptance indicator (i.e., $z^\tau = 1$, if the order is accepted; $z^\tau = 0$, otherwise), $\mathrm{T}^{\tau,a}$ is the updated truck information vector, and $\mathrm{D}^{\tau,a}$ is the updated drone information vector. Further, for any decision epoch $\tau > 0$, we define the reward of taking action a_τ at system state s_τ as $r(s_\tau, a_\tau) = z^\tau$. At the initial decision epoch (i.e., $\tau = 0$), the reward is equal to the total number of initial orders, i.e., $r(s_0, a_0) = |\mathcal{A}^0|$ (since all orders available at the beginning of the day will be delivered).

3.2.1. Illustrative example

An illustrative example is shown in Fig. 1, considering two drones with a load capacity of $Q = 2$ orders. Fig. 1(a) shows the system state at τ: $\mathcal{X}^\tau = (1, 2, 3, 4, 5, 7, 8, 9, 0)$ is the current truck route, the gray nodes represent the orders already loaded onto the truck ($\mathcal{L}^\tau$), and the white nodes are the orders available at the depot ($\mathcal{A}^\tau$). The dotted white node is the newly available order (i.e., $\vartheta^\tau = 6$). Regarding the drone information, the gray dashed lines represent the planned sortie for Drone 1, while the black dotted lines represent both the current and the planned sorties for Drone 2. Mathematically:

- **Drone 1**: $\mathrm{D}_1^\tau = \left(\lambda^\tau,\ \emptyset,\ (\{3,4\}, 2, \xi_{1(1)}^\tau)\right)$, which means that Drone 1 is available at the depot, there are no orders onboard it, and a sortie to send Orders 3 and 4 to Node 2 is planned to start at time $\xi_{1(1)}^\tau$.
- **Drone 2**: $\mathrm{D}_2^\tau = \left(\alpha_2^\tau,\ \emptyset,\ (\{8,9\}, 8, \xi_{2(1)}^\tau)\right)$, which means that Drone 2 will be available at the depot at time α_2^τ, there are no orders onboard it (since the drone is returning to the depot after resupplying the truck in a previous sortie), and it will start a new sortie at time $\xi_{2(1)}^\tau$ to send Orders 8 and 9 to the truck at Node 8.

3.2.2. Transitions from pre- to post-decision state

Assuming that the newly available order is accepted (i.e., $z^\tau = 1$), Fig. 1(b) shows the post-decision state, denoted as $s_{\tau,a}$ (each component of the post-decision state has a superscript a, e.g., $\mathcal{A}^{\tau,a}$ refers to the set of accepted orders that are available at the depot under action a_τ). Note that the planned drone sorties are adjusted to send the new order to the truck. In particular, the components of the system state that are updated are described in Table 2. Note that the collection $\left\{\mathcal{R}_{k(i)}^{\tau,a}\right\}_{k\in K, i\in \mathcal{I}_k^{\tau,a}} = \{\{3,4\},\{9\},\{6,8\}\}$ is a partition of $\mathcal{A}^{\tau,a} = \{3,4,6,8,9\}$, which means that each available order at the depot is included in exactly one drone sortie.

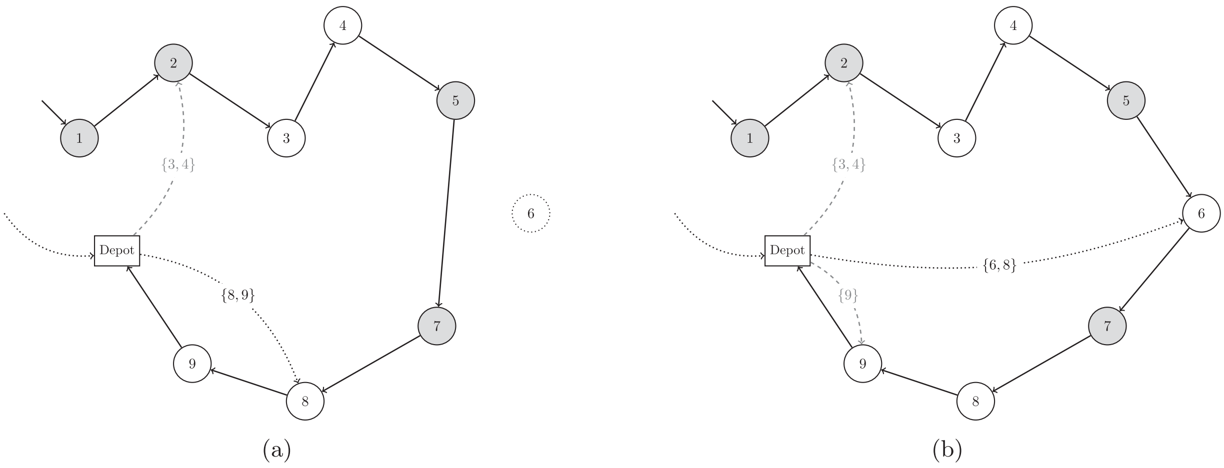


Fig. 1. Example of an action. (a) System state at decision epoch τ. (b) Post-decision state at decision epoch τ.

Table 2
Components of the system state that are updated (see Fig. 1).

Component	Pre-decision state s_τ	Post-decision state $s_{\tau,a}$
Nodes visited by the truck route	$\mathcal{N}^\tau = \{0,1,2,3,4,5,7,8,9\}$	$\mathcal{N}^{\tau,a} = \{0,1,2,3,4,5,6,7,8,9\}$
Truck route	$\chi^\tau = (1,2,3,4,5,7,8,9,0)$	$\chi^{\tau,a} = (1,2,3,4,5,6,7,8,9,0)$
Set of orders available at the depot	$\mathcal{A}^\tau = \{3,4,8,9\}$	$\mathcal{A}^{\tau,a} = \{3,4,6,8,9\}$
Planned schedule for Drone 1	$\mathrm{P}_1^\tau = \left(\{3,4\},2,\xi^\tau_{1(1)}\right)$	$\mathrm{P}_1^{\tau,a} = \left((\{3,4\},2,\xi^{\tau,a}_{1(1)}),(\{9\},9,\xi^{\tau,a}_{1(2)})\right)$
Planned schedule for Drone 2	$\mathrm{P}_2^\tau = \left(\{8,9\},8,\xi^\tau_{2(1)}\right)$	$\mathrm{P}_2^{\tau,a} = \left(\{6,8\},6,\xi^{\tau,a}_{2(1)}\right)$

3.2.3. Set of feasible actions

Let us introduce the additional notation listed in Table 3 before presenting the set of feasible actions. In particular, assuming that the drone flight endurance is large enough, the set of possible meeting locations in Fig. 1 for the first planned sortie of Drone 1 is $\mathcal{M}^{\tau,a}_{1(1)} = \{1,2,3\}$, meaning that Orders 3 and 4 must be transferred to the truck at Nodes 1, 2, or 3 (i.e., before or at the time the truck visits the location associated with Order 3).

Table 3
Additional notation for the action space.

Symbol	Description
H	Delivery horizon (i.e., the time when the truck must return to the depot).
$\mathcal{M}^{\tau,a}_{k(i)}$	Set of all possible meeting locations where drone $k \in K$ can send orders in $\mathcal{R}^{\tau,a}_{k(i)}$ according to the drone flight endurance.
$\mathbb{I}(n)$	Indicator function that takes the value 1 if a drone that will resupply the truck at node $n \in \mathcal{N}^{\tau,a}$, and 0 otherwise (i.e., $\mathbb{I}(n) = 1$ if and only if $n \in \{m^{\tau,a}_{k(i)}\}_{k\in K, i\in \mathcal{I}^{\tau,a}_k}$).
δ	Time the truck needs to receive a drone en route.
T_n	Truck departure time from node $n \in \mathcal{N}^{\tau,a}$.
$t_{n,n'}$	Truck travel time between nodes n and n', with $n, n' \in \mathcal{N}^{\tau,a}$.
d_n	Drone flying time between the depot and node $n \in \mathcal{N}^{\tau,a}$.

Now, given a system state s_τ, the set of all feasible actions $\mathbb{A}^\tau(s_\tau)$ is formally defined as follows:

$$\mathbb{A}^\tau(s_\tau) := \Bigg\{ a_\tau :$$

$$\chi^{\tau,a} : \{1,\dots,|\mathcal{N}^{\tau,a}|\} \to \mathcal{N}^{\tau,a} \text{ is a bijection with } \chi^{\tau,a}_1 = \chi^\tau_1 \text{ and } \chi^{\tau,a}_{|\mathcal{N}^{\tau,a}|} = 0, \tag{1}$$

$$\mathcal{A}^{\tau,a} = \bigcup_{k\in K} \bigcup_{i\in \mathcal{I}^{\tau,a}_k} \mathcal{R}^{\tau,a}_{k(i)}, \tag{2}$$

$$\mathcal{R}^{\tau,a}_{k(i)} \cap \mathcal{R}^{\tau,a}_{k'(j)} = \emptyset, \qquad \forall k \in K, k' \in K\backslash\{k\}, i \in \mathcal{I}^{\tau,a}_k, j \in \mathcal{I}^{\tau,a}_{k'}, \tag{3}$$

$$\mathcal{R}^{\tau,a}_{k(i)} \cap \mathcal{R}^{\tau,a}_{k(j)} = \emptyset, \qquad \forall k \in K, i \in \mathcal{I}^{\tau,a}_k, j \in \mathcal{I}^{\tau,a}_k\backslash\{i\}, \tag{4}$$

$$z^\tau = 1 \;\Leftrightarrow\; \exists k \in K, i \in \mathcal{I}^{\tau,a}_k, \vartheta^\tau \in \mathcal{R}^{\tau,a}_{k(i)}, \tag{5}$$

$$\left|\mathcal{R}^{\tau,a}_{k(i)}\right| \le Q, \qquad \forall k \in K, i \in \mathcal{I}^{\tau,a}_k, \tag{6}$$

$$m^{\tau,a}_{k(i)} \in \mathcal{M}^{\tau,a}_{k(i)}, \qquad \forall k \in K, i \in \mathcal{I}^{\tau,a}_k, \tag{7}$$

$$T_{\chi^{\tau,a}_1} \geq \lambda^\tau + \delta \cdot \mathbb{I}(\chi^{\tau,a}_1), \tag{8}$$

$$T_{\chi^{\tau,a}_i} \geq T_{\chi^{\tau,a}_{i-1}} + t_{\chi^{\tau,a}_{i-1},\chi^{\tau,a}_i} + \delta \cdot \mathbb{I}(\chi^{\tau,a}_i), \qquad \forall i \in \{2, \dots, |\mathcal{N}^{\tau,a}|\}, \tag{9}$$

$$\xi^{\tau,a}_{k(1)} \geq \alpha^{\tau,a}_k, \qquad \forall k \in K, \tag{10}$$

$$\xi^{\tau,a}_{k(i)} \geq T_{m^{\tau,a}_{k(i-1)}} + d_{m^{\tau,a}_{k(i-1)}}, \qquad \forall k \in K, i \in \mathcal{I}^{\tau,a}_k \backslash \{1\}, \tag{11}$$

$$T_{m^{\tau,a}_{k(i)}} \geq \xi^{\tau,a}_{k(i)} + d_{m^{\tau,a}_{k(i)}} + \delta, \qquad \forall k \in K, i \in \mathcal{I}^{\tau,a}_k, \tag{12}$$

$$\left. T_{\chi^{\tau,a}_{|\mathcal{N}^{\tau,a}|}} \leq H. \right\} \tag{13}$$

Constraint (1) ensures the truck follows a feasible route to deliver all loaded and available orders and then returns to the depot. Constraints (2)–(4) force each available order at the depot to be included in exactly one drone sortie. Constraint (5) ensures the newly available order ϑ^τ to be included in one drone sortie if and only if the order is accepted (i.e., $z^\tau = 1$). The maximum number of orders to be resupplied in each planned drone sortie is limited in Constraints (6). Constraints (7) ensure that each drone resupply occurs at one of the possible meeting locations. Constraints (8)–(12) define the truck departure time from each node. Notably, Constraints (10)–(12) ensure that each drone may start a new sortie as long as it has returned to the depot from a previous sortie. Constraints (12) force drones to arrive at meeting locations at the same time or after the truck arrives (i.e., drones are not allowed to wait for the truck since their dispatch from the depot can always be delayed to arrive exactly when the truck arrives at the meeting locations). Finally, Constraint (13) ensures that the truck returns to the depot at or before the end of the horizon.

Note that the definition of $\mathbb{A}^\tau(s_\tau)$ does not impose any constraints related to the situation where a drone is flying towards the truck when the decision epoch happens. Under this situation, it is reasonable to assume that a drone cannot be diverted once dispatched, which means fixing the truck route until receiving the drone (thus ensuring that the truck will arrive at the rendezvous location on time). The latter can be incorporated in Constraints (1) by fixing the values of the first ℓ elements, where ℓ is the index of the position at which the last in-transit drone will arrive.

3.3. Transitions from post- to pre-decision state

Since new orders arrive dynamically throughout the day, the time when decision epoch $\tau+1$ will occur is unknown. The transition to the new system state is defined by a realization of exogenous information $\omega_{\tau+1}$. This realization reveals the time at which the new order $\vartheta^{\tau+1}$ arrives (denoted as $\lambda^{\tau+1}$), and also defines two additional sets: the set of orders delivered between decision epochs τ and $\tau+1$, $\mathcal{F}^\tau$, and the set of orders resupplied between decision epochs τ and $\tau+1$, $\mathcal{S}^\tau$.

Based on the post-decision state $s_{\tau,a}$ and the realized information $\omega_{\tau+1}$, the components of the system state are updated as follows:

- All resupplied and onboard orders are removed from the set of accepted orders that are available at the depot:

$$\mathcal{A}^{\tau+1} = \mathcal{A}^{\tau,a} \backslash \left(\mathcal{S}^\tau \cup \mathcal{O}^{\tau+1}\right)$$

- The truck route $\chi^{\tau,a}$ is updated simply by removing all orders in $\mathcal{F}^\tau$.
- All delivered orders are removed from the set of loaded orders as follows:

$$\mathcal{L}^{\tau+1} = \left(\mathcal{L}^{\tau,a} \cup \mathcal{S}^\tau\right) \backslash \mathcal{F}^\tau$$

- The arrival time $\alpha^{\tau+1}_k$ and the set of onboard orders $\mathcal{O}^{\tau+1}_k$ are updated to reflect the current status of drone $k \in K$.
- For each drone $k \in K$, each tuple $\mathrm{P}^{\tau,a}_{k(i)}$ whose orders has been already transferred to the truck (i.e., $\mathrm{R}^{\tau,a}_{k(i)} \subseteq \mathcal{S}^\tau$) is removed.

3.3.1. Example of the post- to pre-decision state transitions

Returning to our previous example, assume that the action shown in Fig. 1(b) is taken. Assume now that order $\vartheta^{\tau+1} = 10$ arrives at the depot at time $\lambda^{\tau+1}$. The system state at decision epoch $\tau+1$ is presented in Fig. 2. This figure shows that:

- The truck is currently traveling towards Node 4.
- Drone 1 is at the depot (i.e., $\mathcal{O}^{\tau+1}_1 = \emptyset$), while Drone 2 is currently flying to resupply Orders 6 and 8 to the truck at Node 6 (i.e., $\mathcal{O}^{\tau+1}_2 = \{6, 8\}$).
- The set of orders delivered to customers between decision epochs τ and $\tau+1$ is $\mathcal{F}^\tau = \{1, 2, 3\}$.
- The set of orders resupplied between decision epochs τ and $\tau+1$ is $\mathcal{S}^\tau = \{3, 4\}$.

Based on the above information, the components of the pre-decision state $s_{\tau+1}$ are described in Table 4. Note that $\mathcal{L}^{\tau+1} = (\mathcal{L}^{\tau,a} \cup \mathcal{S}^\tau) \backslash \mathcal{F}^\tau$, $\mathcal{O}^{\tau+1} = \mathcal{O}^{\tau+1}_1 \cup \mathcal{O}^{\tau+1}_2$, and $\mathcal{A}^{\tau+1} = \mathcal{A}^{\tau,a} \backslash (\mathcal{S}^\tau \cup \mathcal{O}^{\tau+1})$.

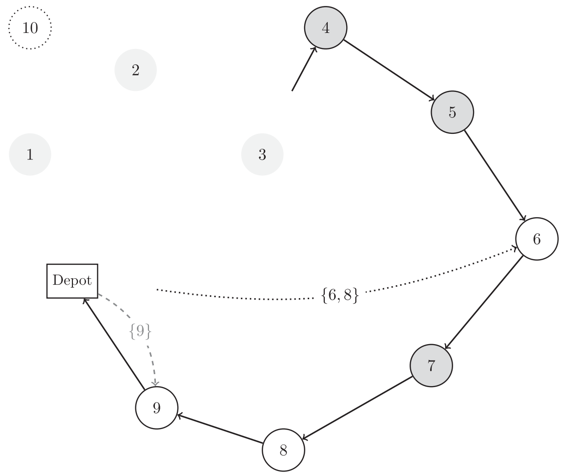


Fig. 2. System state at $\tau + 1$.

Table 4
Components of the pre-decision state at decision epoch $\tau + 1$ (see Table 2 and Fig. 2).

Component	Post-decision state $s_{\tau,a}$	Pre-decision state $s_{\tau+1}$
Truck route	$\mathcal{X}^{\tau,a} = (1,2,3,4,5,6,7,8,9,0)$	$\mathcal{X}^{\tau+1} = (4,5,6,7,8,9,0)$
Orders loaded onto the truck	$\mathcal{L}^{\tau,a} = \{1,2,5,7\}$	$\mathcal{L}^{\tau+1} = \{4,5,7\}$
Orders loaded onto the drones	$\mathcal{O}^{\tau,a} = \emptyset$	$\mathcal{O}^{\tau+1} = \{6,8\}$
Orders available at the depot	$\mathcal{A}^{\tau,a} = \{3,4,6,8,9\}$	$\mathcal{A}^{\tau+1} = \{9\}$
Vector information of Drone 1	$\mathrm{D}_1^{\tau,a} = \left(\lambda^{\tau,a}, \emptyset, \mathrm{P}_1^{\tau,a}\right)$	$\mathrm{D}_1^{\tau+1} = \left(\lambda^{\tau+1}, \emptyset, (\{9\}, 9, \xi^{\tau,a}_{1(1)})\right)$
Vector information of Drone 2	$\mathrm{D}_2^{\tau,a} = \left(\alpha_2^{\tau,a}, \emptyset, \mathrm{P}_2^{\tau,a}\right)$	$\mathrm{D}_2^{\tau+1} = \left(\alpha_2^{\tau,a}, \{6,8\}, -\right)$

3.4. Objective function

Let $\mathcal{T}$ be a random variable representing the number of decision epochs. Furthermore, let μ_τ be a function that maps each state s_τ into an action $\mu_\tau(s_\tau) = a_\tau$. We aim to find a sequence of functions $\pi = \left(\mu_\tau^\pi\right)_{\tau=0}^{\mathcal{T}}$ (a so-called policy) that maximizes the expected number of orders delivered within the working hours of the depot, as presented below:

$$\pi^* \in \arg\max_{\pi \in \Pi} \ \mathbb{E}\left[\sum_{\tau=0}^{\mathcal{T}} r\left(s_\tau, \mu_\tau^\pi(s_\tau)\right) \ \Big| \ s_0\right]. \tag{DTSP-DR}$$

4. Perfect-information upper bound

An upper bound to Problem (DTSP-DR) can be computed through its static counterpart, where the total number of customers requesting SDD and their order arrival times are known before the operation starts. We term this counterpart the Orienteering Problem with Release Dates and Drone Resupply (OPRD-DR), which is an extension of the classic Orienteering Problem (OP) that considers release dates and multiple resupply drones to support the delivery operation (see Vansteenwegen et al. (2011) for a survey on OPs). To our knowledge, the OPRD-DR has not been studied to date, so the following MILP formulation is new to the literature.

4.1. Mathematical formulation

Let $\mathcal{A}^0$ be the set of orders available at the depot at the beginning of the day. Define $\boldsymbol{\omega}$ as a random variable that defines both the set of dynamic orders and their arrival times. Given a realization $\tilde{\boldsymbol{\omega}}$ of $\boldsymbol{\omega}$, let $\mathcal{E}(\tilde{\boldsymbol{\omega}})$ be the set of all realized dynamic orders and $N := \mathcal{A}^0 \cup \mathcal{E}(\tilde{\boldsymbol{\omega}}) = \{1, \dots, n\}$ (for the sake of readability, we omit an index $\tilde{\boldsymbol{\omega}}$ in the following). Furthermore, let K be the set of resupply drones and $D \subseteq N$ be the set of customer locations where drones can resupply the truck according to their flight endurance. The depot is denoted with the subindices 0 and $n+1$. Let $D_0 = D \cup \{0\}$, $D_{n+1} = D \cup \{n+1\}$, and $N_{0,n+1} = N \cup \{0, n+1\}$ be the sets of nodes that include the depot. The OPRD-DR is defined on the complete directed graph $G = (N_{0,n+1}, E)$, where E is the set of all edges between the nodes. The notation of the MILP formulation of the OPRD-DR is described in Table 5.

Table 5
Notation for the MILP model of the OPRD-DR.

Symbol	Description
Parameters	
r_i	Arrival time of order $i \in N$.
t_{ij}	Truck travel time associated with arc $(i,j) \in E$.
d_i	Drone flying time between the depot and customer $i \in N$.
δ	Time the truck driver needs to receive and unload a drone.
Q	Drone maximum load capacity per sortie.
H	Delivery horizon.
M	A sufficiently large constant.
Decision variables	
z_i	1, if order $i \in N$ is delivered. 0, otherwise.
x_{ij}	1, if the truck visits node j after node i, with $(i,j) \in E$. 0, otherwise.
w^k_{ij}	1, if drone $k \in K$ visits customer $i \in N$, returns to the depot, and starts a new sortie towards customer $j \in N$, with $i \neq j$. 0, otherwise.
w^k_{0j}	1, if drone $k \in K$ performs the first sortie to customer $j \in N$. 0, otherwise.
$w^k_{i,n+1}$	1, if drone $k \in K$ performs the last sortie to customer $i \in N$. 0, otherwise.
u^k_i	1, if node $i \in D$ is visited by drone $k \in K$. 0, otherwise.
u^k_0	1, if drone $k \in K$ resupplies the truck at least once. 0, otherwise.
y^k_{ij}	1, if order $j \in N$ is loaded onto a truck at customer $i \in D$ via drone $k \in K$. 0, otherwise.
γ_i	1, if order $i \in N$ is loaded onto the truck at the depot. 0, otherwise.
T_i	Time when the truck departs node $i \in N_{0,n+1}$.

4.1.1. MILP model

The MILP model is presented below.

$$\mathcal{Z}(\tilde{\boldsymbol{\omega}}) \;:=\; \max \sum_{i \in N} z_i, \tag{OPRD-DR}$$

subject to:

- Truck and drone routing:

$$\sum_{j \in N} x_{0j} = 1, \tag{14}$$
$$\sum_{i \in N} x_{i,n+1} = 1, \tag{15}$$
$$\sum_{j:(i,j)\in E} x_{ij} = z_i, \qquad \forall i \in N, \tag{16}$$
$$\sum_{i:(i,j)\in E} x_{ij} = z_j, \qquad \forall j \in N, \tag{17}$$
$$z_i = 1, \qquad \forall i \in \mathcal{A}^0, \tag{18}$$
$$\sum_{k \in K} u^k_i \leq 1, \qquad \forall i \in D, \tag{19}$$
$$u^k_i \leq u^k_0, \qquad \forall k \in K, i \in D, \tag{20}$$
$$\sum_{j \in D} w^k_{0j} = u^k_0, \qquad \forall k \in K, \tag{21}$$
$$\sum_{i \in D} w^k_{i,n+1} = u^k_0, \qquad \forall k \in K, \tag{22}$$
$$\sum_{j \in D_{n+1}\setminus\{i\}} w^k_{ij} = u^k_i, \qquad \forall k \in K, i \in D, \tag{23}$$
$$\sum_{i \in D_0\setminus\{j\}} w^k_{ij} = u^k_j, \qquad \forall k \in K, j \in D, \tag{24}$$

- Loading and linking constraints:

$$\sum_{j \in N} y^k_{ij} \leq Q \cdot u^k_i, \qquad \forall k \in K, i \in D, \tag{25}$$
$$\sum_{j \in N} y^k_{ij} \geq u^k_i, \qquad \forall k \in K, i \in D, \tag{26}$$

$$\sum_{k\in K}\sum_{i\in D} y_{ij}^k + \gamma_j = z_j, \qquad \forall j \in N, \tag{27}$$

$$\sum_{k\in K} u_i^k \leq z_i, \qquad \forall i \in D, \tag{28}$$

- Synchronization and timing constraints:

$$T_j \geq T_i + t_{ij} - M \cdot \left(1 - \sum_{k\in K} y_{ij}^k\right), \qquad \forall i \in N, j \in D\backslash\{i\}, \tag{29}$$

$$T_j \geq T_i + t_{ij} - M \cdot (1 - x_{ij}), \qquad \forall i \in N_0, j \in N_{n+1}\backslash(D \cup \{i\}), \tag{30}$$

$$T_j \geq T_i + t_{ij} + \delta \cdot \left(\sum_{k\in K} u_j^k\right) - M \cdot (1 - x_{ij}), \qquad \forall i \in N_0, j \in D\backslash\{i\}, \tag{31}$$

$$T_j \geq T_i + d_i + d_j + \delta - M \cdot \left(1 - \sum_{k\in K} w_{ij}^k\right), \qquad \forall i \in D, j \in D\backslash\{i\}, \tag{32}$$

$$T_i \geq (r_j + d_i + \delta) \cdot \left(\sum_{k\in K} y_{ij}^k\right), \qquad \forall i \in D, j \in D, \tag{33}$$

$$T_0 \geq r_i \cdot \gamma_i, \qquad \forall i \in N, \tag{34}$$

$$T_{n+1} \leq H, \tag{35}$$

- Non-negativity and integrality constraints:

$$x_{ij} \in \{0,1\}, \qquad \forall (i,j) \in E, \tag{36}$$

$$z_i, \gamma_i \in \{0,1\}, \qquad \forall i \in N, \tag{37}$$

$$w_{ij}^k \in \{0,1\}, \qquad \forall i \in D_0, j \in D_{n+1}\backslash\{i\}, k \in K, \tag{38}$$

$$u_i^k \in \{0,1\}, \qquad \forall i \in D_0, k \in K, \tag{39}$$

$$y_{ij}^k \in \{0,1\}, \qquad \forall i \in D, j \in N, k \in K, \tag{40}$$

$$T_i \geq 0, \qquad \forall i \in N_{0,n+1}. \tag{41}$$

In Problem (OPRD-DR), the objective function maximizes the total number of orders delivered. Constraints (14)–(17) are the classic flow balance constraints. Constraints (18) ensure delivery of each order that is available at the depot at the beginning of the day. Constraints (19) restrict at most one drone resupply at each customer location. Constraints (20) indicate that drone $k \in K$ can visit a customer as long as it is used to resupply the truck at least once (i.e., $u_0^k = 1$). Constraints (21)–(24) define drone flow conservation for each node. The maximum number of orders resupplied at each drone sortie is limited in Constraints (25). Constraints (26) state a drone must resupply at least one order on each sortie. Constraints (27) ensure that the order of each customer served is loaded. Together with Constraints (27), Constraints (28) link the drone and truck variables. As a group, Constraints (29)–(33) allow computing the departure time at each customer, as well as synchronizing the drone sorties with the truck route. Specifically, Constraints (29) ensure that the departure time at customer $j \in N$ must be greater than the departure time at customer $i \in N$ if the order of customer $j \in N$ is loaded onto the truck at customer $i \in N$ via a drone. Constraints (30) and (31) are for subtour elimination. Constraints (32) and (33) model that a drone can start a new sortie as long as it has returned to the depot (from a previous sortie), and all orders to be resupplied have been released. The departure time of the truck from the depot is bounded in Constraints (34). Constraint (35) forces the truck to return to the depot before the delivery horizon expires. Finally, the domain of each decision variable is specified in Constraints (36)–(41).

Extending Pina-Pardo et al. (2021), we derive the following valid inequalities to strengthen the above formulation (Table A.10 in Appendix shows the effectiveness of these inequalities in reducing both run-times and optimality gaps):

$$T_{n+1} \geq T_i + t_{i,n+1} - M \cdot (1 - z_i), \qquad \forall i \in N, \tag{42}$$

$$T_i \geq r_i + \min\{d_i + \delta, t_{0i}\}, \qquad \forall i \in N, \tag{43}$$

$$T_{n+1} \geq T_0 + \sum_{(i,j)\in E} t_{ij} x_{ij} + \sum_{i\in D} \epsilon_i, \tag{44}$$

$$\epsilon_j \geq T_j - (T_i + t_{ij}) - M \cdot (1 - x_{ij}), \qquad \forall j \in D, i \in N_0\backslash\{j\}, \tag{45}$$

$$\epsilon_j \leq M \cdot \sum_{k\in K} u_j^k, \qquad \forall j \in D, \tag{46}$$

$$\epsilon_j \geq 0, \qquad \forall j \in D. \tag{47}$$

Constraints (42) impose that the time when the truck returns to the depot is bounded by its departure time from any served customer plus the truck travel time from that customer to the depot. Constraints (43) define a lower bound for the departure time at each customer. Constraint (44) imposes a second lower bound on the completion time, defined by the time when the truck departs the depot, plus the total truck travel time, plus any possible waiting time of the truck for the drones. These waiting times are computed by decision variables ϵ through Constraints (45)–(47).

4.1.2. Perfect-information upper bound

Given a realization $\tilde{\boldsymbol{\omega}}$ of the random variable $\boldsymbol{\omega}$, the MILP model of Problem (OPRD-DR) provides a valid upper bound to the DTSP-DR. Note this is true even if an optimal solution is not found within a given time limit (in this case, since Problem (OPRD-DR) is a maximization problem, the upper bound given by the MILP solver still provides a valid upper bound to the DTSP-DR). Furthermore, we can estimate an upper bound on the optimal expected value of Problem (DTSP-DR) through computing $\mathbb{E}[\mathcal{Z}(\boldsymbol{\omega})]$ (see, e.g., Klapp et al. (2018)). However, the set of all potential realizations is uncountable in our case, rendering the exact calculation of $\mathbb{E}[\mathcal{Z}(\boldsymbol{\omega})]$ computationally prohibitive. Therefore, following the sample average approximation technique (see, e.g., Ahmed and Shapiro (2002)), we estimate $\mathbb{E}[\mathcal{Z}(\boldsymbol{\omega})]$ by the average of a sample $\{\tilde{\boldsymbol{\omega}}_l\}_{l=1}^L$ of L realizations, as presented in Eq. (48):

$$\mathcal{Z}_{\mathrm{UB}} := \frac{1}{L}\sum_{l=1}^{L} \mathcal{Z}(\tilde{\boldsymbol{\omega}}_l). \tag{48}$$

5. Solution approach

A solution to the DTSP-DR consists of defining an initial truck route and an online policy to dynamically update the truck route and the drone resupply operations based on newly arrived orders. The initial truck route is determined offline and defines a sequence to serve all orders available at the beginning of the day. On the other hand, given the need to respond immediately to customer service requests, the online policy requires quickly determining whether to accept or reject newly arrived orders and adjust truck and drone operations accordingly. To this end, this section describes an efficient online policy for the DTSP-DR based on reoptimization, which combines an insertion heuristic with a fast MILP model to adjust drone resupply operations optimally.

5.1. Overview of the solution approach

Fig. 3 presents a flowchart of the proposed solution approach. At the first decision epoch, we set the initial truck route as an optimal Traveling Salesman Problem (TSP) tour covering all initial orders. Then, every time a new order arrives, we update the system to the next decision state (see Section 3.3). Next, we reoptimize the current distribution plan and accept the newly arrived order according to an order acceptance mechanism (see Section 5.2). In case of acceptance, we implement the reoptimized distribution; that is, we increase the total number of orders accepted and transition the system state from the pre- to the post-decision state (see Section 3.2.2). Note that the system state remains the same in case of rejecting the newly arrived order. The algorithm ends when the truck route cannot accommodate new orders or when an order arrival time exceeds the delivery horizon, whichever occurs first.

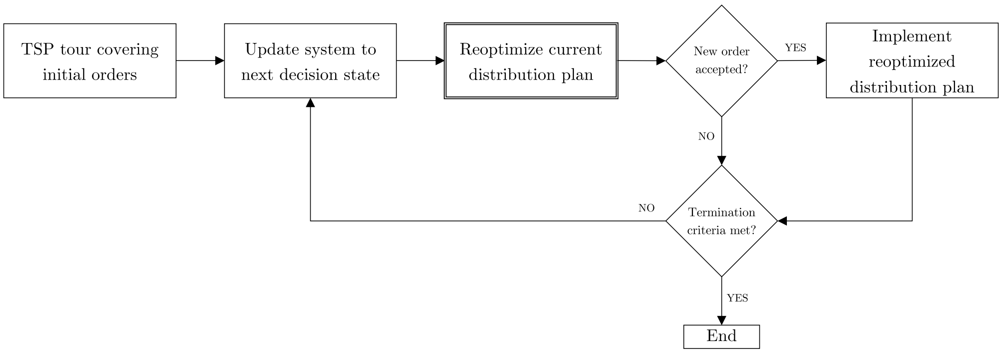


Fig. 3. Solution approach.

5.2. Reoptimization of the current distribution plan

The reoptimization of the current distribution plan works as follows: First, the truck route is updated by inserting the newly arrived order using the well-known Cheapest Insertion technique (Rosenkrantz et al., 1977). This insertion technique guarantees a tight worst-case bound with a ratio equal to 3/2 and runs in linear time (Archetti et al., 2003), making it appropriate to satisfy the tight time frames imposed by real-time decision-making. Furthermore, the resulting route is not disruptive for the driver since the relative position of the rest of the orders does not change when adding a new order to the route (Ulmer and Thomas, 2018; Ulmer et al., 2019; Zhang et al., 2023).

Second, if there exists at least one location on the updated truck route (i.e., the route after inserting the newly arrived order) where a drone can reach the truck based on its flight endurance (otherwise, the order is rejected), we compute the following lower bound on the completion time of the updated truck route:

$$\mathrm{LB}(\chi^{\tau,a}) \;:=\; T_{\chi_1^{\tau,a}} \;+\; \sum_{i=2}^{\eta} t_{\chi_{i-1}^{\tau,a},\chi_i^{\tau,a}} \;+\; \delta \cdot \left\lceil \frac{|\mathcal{A}^\tau|}{Q} \right\rceil, \tag{49}$$

where $\chi^{\tau,a} = (\chi_1^{\tau,a}, \dots, \chi_\eta^{\tau,a})$ is the updated truck route and $T_{\chi_1^{\tau,a}}$ is the time when the truck will depart the first node of this revised tour. Note that, since the term $\lceil |\mathcal{A}^\tau|/Q \rceil$ constitutes a lower bound on the total number of sorties required to resupply all available orders at the depot, $\mathrm{LB}(\chi^{\tau,a})$ is equal to the completion time of the updated truck route if the truck does not have to wait for any drones en route. If $\mathrm{LB}(\chi^{\tau,a})$ is not greater than the delivery deadline (otherwise, the order is rejected), we invoke the Drone Resupply Problem (DRP) to adjust the drone resupply operations such that the completion time of the updated truck route is minimized (see Section 5.3). The newly arrived order is rejected if the optimal DRP value exceeds the delivery deadline.

Third, an order acceptance mechanism accepts the newly arrived order if the time consumption is not greater than a given threshold. This mechanism aims to select orders that are efficient with the current route and potentially decline orders for SDD that alter the current route too radically, even though it would actually be feasible to accept them. We define the time consumption as the increase in completion time resulting from inserting the newly arrived order and updating the drone resupply operations. The order acceptance mechanism is defined as follows:

$$z^\tau \;=\; \begin{cases} 1, & \text{if } C(s_{\tau,a}) - C(s_\tau) \leq \textsc{Threshold}(\tau) \\ 0, & \text{otherwise} \end{cases}, \tag{50}$$

where $C(s_\tau)$ and $C(s_{\tau,a})$ denote the completion time of the distribution plans described by the current system state (s_τ) and the post-decision state ($s_{\tau,a}$) after solving the DRP (see Section 5.3), respectively, and $\textsc{Threshold}(\tau)$ denotes the value of the threshold at decision epoch τ.

In principle, our order acceptance mechanism allows using any threshold to decide whether to accept or reject new orders. For instance, the threshold could be a constant or a number large enough to accept any order that can be accommodated in the current distribution plan, regardless of the time consumption. In this work, we define the threshold as the fraction of the slack time relative to the end of the delivery horizon that can be consumed by accepting a new order. Formally,

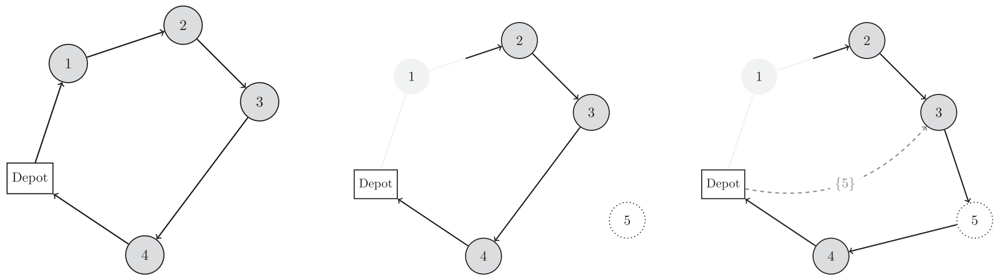
$$\textsc{Threshold}(\tau) := \rho_\tau \cdot \left(H - C(s_\tau)\right), \tag{51}$$

where $\rho_\tau \in [0,1]$ for each decision epoch τ. Note that different policies can be defined depending on the value of ρ_τ. In particular, we define an adaptive policy π^{A} with

$$\rho_\tau^{\mathrm{A}} = \frac{|\mathcal{A}^0| + \tau}{|\mathcal{A}^0| + \mathbb{E}[\mathcal{T}]}, \tag{52}$$

where $\mathbb{E}[\mathcal{T}]$ denotes the expected number of decision epochs (which is equivalent to the expected number of dynamic orders). Note that Eq. (52) allows the threshold to be more restrictive at the beginning of the day and when many potential dynamic orders are still expected to arrive later. For benchmarking purposes, we also define a greedy policy π^{G} with $\rho_\tau^{\mathrm{G}} = 1$, which accepts any order that can be feasibly served regardless of the time consumption.

Fig. 4 shows an illustrative example of our solution approach. At the first decision epoch, the truck is dispatched to visit Customers 1, 2, 3, and 4 through an optimal TSP tour. At the second decision epoch, the order of Customer 5 becomes available for dispatch. This order is inserted into the current truck route through the Cheapest Insertion technique and the DRP is then solved to define the optimal drone resupply operation to transfer the order to the truck.



(a) Optimal TSP tour serving all initial customers. (b) System state at the second decision epoch. (c) Insertion point and drone resupply operation.

Fig. 4. Example of the proposed solution approach at the first and second decision epochs.

5.3. The drone resupply problem

The DRP defines the optimal drone sorties to resupply all accepted orders that are currently at the depot. In the model formulation, if a drone is currently flying towards the truck when the decision epoch occurs, we consider that future drone resupplies must occur after the truck receives that drone, thus ensuring that the truck will arrive at the corresponding rendezvous location on time (see Section 3.2.3).

5.3.1. Model formulation

Let $\chi^{\tau,a} = (\chi_1^{\tau,a}, \dots, \chi_\eta^{\tau,a})$ be the truck route after inserting the newly arrived order and ℓ be the position of $\chi^{\tau,a}$ where the last in-transit drone will meet the truck. If no drone is currently in transit, we define ℓ as the first position of $\chi^{\tau,a}$ where a drone can reach the truck according to its flight endurance. Furthermore, let $\mathcal{J} := \{1, \dots, \lceil |\mathcal{A}^\tau|/Q \rceil\}$ be the set of sorties required to resupply all available orders at the depot. To streamline the MILP formulation, define $\mathcal{I}_j$ as the subset of indices of $\chi^{\tau,a}$ where the jth sortie can take place (which considers the drone flight endurance), $\mathcal{D} := \cup_{j\in\mathcal{J}} \mathcal{I}_j$, and $\mathcal{I} := \{\ell, \dots, \max\{i : i \in \mathcal{D}\}\}$. Note that $\mathcal{I}_1$ starts from position ℓ. Finally, we define $\mathcal{J}_i := \{j \in \mathcal{J} : i \in \mathcal{I}_j\}$ for each $i \in \mathcal{D}$.

The notation of the MILP model of the DRP is listed in Table 6. Note that the indices of the parameters and decision variables now indicate positions in the truck route and not the node indices.

Table 6
Notation for the MILP model of the DRP.

Symbol	Description
Parameters	
$t_{i-1,i}$	Truck travel time between customers visited at positions $i-1$ and i, with $i \in \mathcal{I}\backslash\{\ell\}$.
d_i	Drone flying time between the depot and the node visited at position $i \in \mathcal{D}$.
α_k	Time when drone $k \in K$ will be available at the depot.
$\tilde{T}_\ell$	Time when the truck will be ready to depart the node visited at position ℓ.
δ	Time the truck driver needs to receive and unload a drone en route.
Decision variables	
u_i^k	1, if drone $k \in K$ flies to the node visited at position $i \in \mathcal{D}$ to resupply the truck. 0, otherwise.
y_{ji}	1, if the jth sortie takes place at position $i \in \mathcal{I}_j$. 0, otherwise.
T_i	Time when the truck departs the node visited at position $i \in \mathcal{I}$.

5.3.2. MILP model

Given a fixed truck route $\chi^{\tau,a}$, the MILP model of the DRP is given below.

$$C_{\text{DRP}} := \min T_{|\mathcal{I}|}, \tag{DRP}$$

subject to:

$$\sum_{i\in\mathcal{I}_j} y_{ji} = 1, \quad \forall j \in \mathcal{J}, \tag{53}$$

$$y_{j'i'} \le 1 - y_{ji}, \quad \forall j, j' \in \mathcal{J}, i \in \mathcal{I}_j, i' \in \mathcal{I}_{j'}, j' > j, i' \le i, \tag{54}$$

$$\sum_{j\in\mathcal{J}_i} y_{ji} = \sum_{k\in K} u_i^k, \quad \forall i \in \mathcal{D}, \tag{55}$$

$$T_\ell \ge \tilde{T}_\ell + \delta \cdot \left(\sum_{k\in K} u_\ell^k\right), \tag{56}$$

$$T_i \ge T_{i-1} + t_{i-1,i}, \quad \forall i \in \mathcal{I}\backslash\mathcal{D}, \tag{57}$$

$$T_i \ge T_{i-1} + t_{i-1,i} + \delta \cdot \left(\sum_{k\in K} u_i^k\right), \quad \forall i \in \mathcal{D}\backslash\{\ell\}, \tag{58}$$

$$T_i \ge (\alpha_k + d_i + \delta) \cdot u_i^k, \quad \forall k \in K, i \in \mathcal{D}, \tag{59}$$

$$T_i \ge T_j + (d_j + \delta + d_i) \cdot (u_i^k + u_j^k - 1), \quad \forall k \in K, i \in \mathcal{D}\backslash\{\ell\}, j \in \mathcal{D}, j < i, \tag{60}$$

$$u_i^k \in \{0, 1\}, \quad \forall k \in K, i \in \mathcal{D}, \tag{61}$$

$$y_{ji} \in \{0, 1\}, \quad \forall j \in \mathcal{J}, i \in \mathcal{I}_j, \tag{62}$$

$$T_i \ge 0, \quad \forall i \in \mathcal{I}. \tag{63}$$

In Problem (DRP), the objective function minimizes the total time to reach customer $\chi_{|\mathcal{I}|}^{\tau,a}$. Constraints (53) ensure the definition of all required sorties to resupply all orders available at the depot. Constraints (54) are for symmetry breaking, ensuring that sortie j' be performed after sortie j, for $j, j' \in \mathcal{J}$ such that $j' > j$. Constraints (55) link decision variables y_{ji} and u_i^k. Note that Constraints (54) and (55) together force at most one resupply operation to occur at each position. Constraints (56)–(60) define the departure time from each node. Specifically, Constraints (59) and (60) establish that a drone can start a new sortie as long as it has returned to the depot (from a previous sortie). Finally, the domain of each decision variable is specified in Constraints (61)–(63).

To strengthen the above formulation, we also introduce the following valid inequality derived from Eq. (49), which provides a valid lower bound to Problem (DRP):

$$T_{|\mathcal{I}|} \ge \tilde{T}_\ell + \sum_{i=\ell+1}^{|\mathcal{I}|} t_{i-1,i} + \delta \cdot |\mathcal{J}|. \tag{64}$$

Finally, the completion time of the post-decision state $s_{\tau,a}$ is defined as follows (see Eq. (50)):

$$C(s_{\tau,a}) := C_{\mathrm{DRP}} + \sum_{i=|\mathcal{I}|+1}^{\eta} t_{i-1,i}. \tag{65}$$

6. Numerical experiments

This section describes and reports our computational experiments. The solution approach was programmed in Java on a workstation Intel(R) Xeon(R) Gold 5118 CPU @2.30 GHz (12 cores) with 64 GB RAM, using CPLEX 12.8, when necessary, as a MILP solver.

6.1. Instances and parameter values

To evaluate the performance of the proposed solution approach and derive managerial insights, we adapted the set of instances proposed by Archetti et al. (2018). These instances are based on Solomon's instances C101, C201, R101, and RC101 (Solomon, 1987). For a given number of customers (n), Archetti et al. (2018) uses the first $n+1$ nodes from each Solomon's instance, assuming that the first node represents the depot. The authors compute the optimal TSP value (z_{TSP}) of serving all customers (without release times and delivery deadlines) and then generate uniformly random release times in the time interval $[0, \beta \cdot z_{\mathrm{TSP}}]$, where β is a scalar that defines the dispersion of release times. In our case, we also introduce the parameter DOD to define the degree of dynamism – i.e., the fraction of orders that are dynamic (Larsen et al., 2002). For instance, DOD = 0 denotes the case where all orders are available to dispatch at the beginning of the day. Similar to Ulmer et al. (2018), we use DOD $\in \{0.50, 0.75\}$. Additionally, we define the delivery horizon as $H = \zeta \cdot (\beta \cdot z_{\mathrm{TSP}})$, where ζ is a factor that controls the length of the horizon with respect to the upper bound of the time interval used to generate order release times. In our experiments, we use $\beta = 1.0$ and $\zeta \in \{1.00, 1.25, 1.50\}$ to establish the delivery horizons. For each value of DOD and customer distribution, we create ten different instances by randomly setting $\lfloor n \cdot (1 - \mathrm{DOD}) \rfloor$ orders as initial orders (i.e., their release times are set to zero). Release times of non-initial (i.e., dynamic) orders remain the same as defined by Archetti et al. (2018). An online repository of the test instances and results can be found on GitHub at https://github.com/jcpinap/DTSP-DR.

For vehicle parameters, we consider a truck speed of 1 distance unit per time unit and a drone speed of twice the truck speed, assuming that vehicles travel the Euclidean distance between nodes (truck parameters are consistent with those used by Archetti et al. (2018) to generate their instances). We also assume that the drone flight endurance is enough to reach and resupply the truck at any customer location. We set the maximum drone payload capacity at $Q = 4$ orders. Considering that most orders delivered by Amazon weigh less than 2.3 kg (Poikonen and Golden, 2020), this payload capacity is a conservative estimate for existing commercial drones such as the Avidrone 210TL, which can carry up to 25 kg (i.e., roughly ten packages per sortie) (Avidrone Aerospace, 2021). Finally, for simplicity and without loss of generality, we set $\delta = 0$. The big-M value is $M = H$ (since the departure time from each visited node must be less than the delivery horizon).

6.2. Comparison with perfect-information scenario

We begin by comparing the results of our solution approach with the adaptive policy π^{A} to the perfect-information scenario. To do so, we compute the following metrics:

- Fill rate: Percentage of total (i.e., initial and dynamic) orders delivered.
- Perfect-information gap: Relative percentage difference between the objective value of the adaptive policy and the objective value of Problem (OPRD-DR). Formally, for a given instance, the perfect-information gap is computed as follows:

$$\mathrm{Gap} = \frac{\mathcal{Z}_{\mathrm{PI}} - \mathcal{Z}_{\mathrm{A}}}{\mathcal{Z}_{\mathrm{PI}}},$$

  where $\mathcal{Z}_{\mathrm{PI}}$ is the optimal value of Problem (OPRD-DR) and $\mathcal{Z}_{\mathrm{A}}$ is the value of the adaptive policy.

Considering the R101 customer distribution, $\zeta = 1$, and two drones with a load capacity of four orders each, Fig. 5 presents the distribution of the perfect-information gaps over 10 instances for each combination of number of customers and degree of dynamism. Overall, gaps increase as the percentage of dynamic orders increases. For example, for instances of 15 customers, our solution approach has a median decrease of 7.7% over the perfect-information counterpart when DOD = 50%, while a median decrease of 16.7% when DOD = 75%. This behavior is expected and reveals the challenge of deciding on the fly whether to accept newly arrived orders as the degree of dynamism increases.

To provide some further insight into the computational speed and efficacy of the perfect-information model, Table 7 reports the average results of the MILP model (14)–(47), including run times and optimality gaps, over 10 instances, considering $n \in \{10, 15, 20, 25, 30\}$ customers, the R101 customer distribution, two drones with a load capacity of four orders each, $\zeta = 1.0$, and a maximum run-time of two hours. Fill rates were computed using the objective value obtained by CPLEX within the maximum run time. As expected, as instances become larger and more dynamic, the perfect-information MILP model (14)–(47) becomes intractable.

Now, we explore the impact of the delivery horizon on the fill rates. Considering DOD = 75% and $\zeta \in \{1.00, 1.25, 1.50\}$, Fig. 6 shows the average fill rates reached by our solution approach as well as for the perfect-information counterpart. For a given number

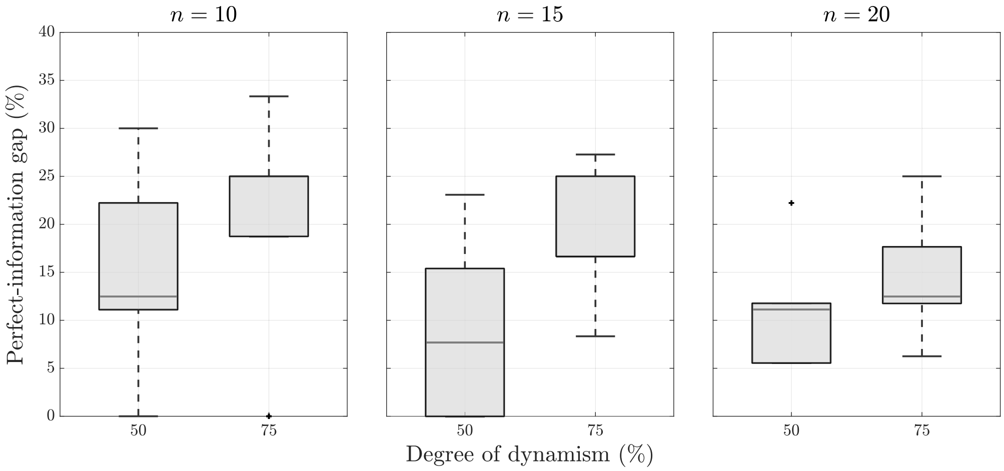


Fig. 5. Distribution of the perfect-information gaps (%) for different instance sizes (n) and degrees of dynamism (DOD). For $n = 10$ and DOD = 75, the median perfect-information gap is 25%, while 16.7% for $n = 15$ and DOD = 75.

of customers, fill rates get closer to perfect information as the delivery horizon increases. This shows that our solution approach performs better when there is more flexibility to integrate newly arrived orders into the ongoing distribution plan. Moreover, $\zeta = 1.50$ seems too relaxed for these instances; the dispatcher can deliver (nearly) all orders regardless of the information at hand when making decisions. From a managerial perspective, this type of analysis can help better define the operational horizon based on the expected number of customer requests. We refer the reader to Fig. A.9 of Appendix for further details on the fill rates achieved for $\zeta = 1$.

Table 7
Average results of the MILP model (14)–(47) over 10 instances.

n	DOD (%)	Fill rate (%)	Run-time (s)	Optimality gap (%)
10	50	88.0	0.1	0.0
	75	81.1	0.2	0.0
15	50	86.7	1.9	0.0
	75	79.3	4.0	0.0
20	50	89.0	43.9	0.0
	75	80.5	651.4	0.0
25	50	86.0	4647.2	4.5
	75	78.0	5178.9	6.9
30	50	83.7	5910.0	10.6
	75	72.0	7214.4	27.9

6.3. Large instances and managerial insights

We now investigate realistic-sized instances of 100 customers. We begin by comparing the performance of our adaptive policy π^{A} to the greedy policy π^{G}, where any order that can be feasibly served is accepted, regardless of the time consumption (see Section 5.2). We then show the value of drone resupply over a more conventional system where resupply is performed by another truck. Finally, we report the decision times of our online policy.

Value of the adaptive policy. Table 8 reports the average fill rates over 10 instances for each of Solomon's customer distributions, considering two drones with a load capacity of four orders each. Results show that our adaptive policy outperforms the greedy policy, especially when customers are clustered. However, fill rates decrease for both policies as customers are more clustered (i.e., C101 and C201 customer distributions). These results are likely because the truck needs to return to the same cluster multiple times during the delivery horizon, and integrating orders into the ongoing distribution route from customers belonging to different clusters can make for inefficient routes.

Interestingly, Table 8 shows that fill rates remain at 68% for the R101 customer distribution, which is similar to the results reported in Fig. 6. We thus conjecture that our solution approach may behave similarly regardless of the number of customers (n), which is appealing considering that the optimal TSP value (the delivery horizon used in Table 8) grows proportionately to the square root of n (i.e., the delivery horizon becomes increasingly narrow as n increases) (Daganzo, 1984).

To further analyze the two policies' behavior throughout the delivery horizon, Fig. 7 presents the average order acceptance rate over time, which is defined as the number of orders accepted over the total number of orders received so far. Unsurprisingly, the

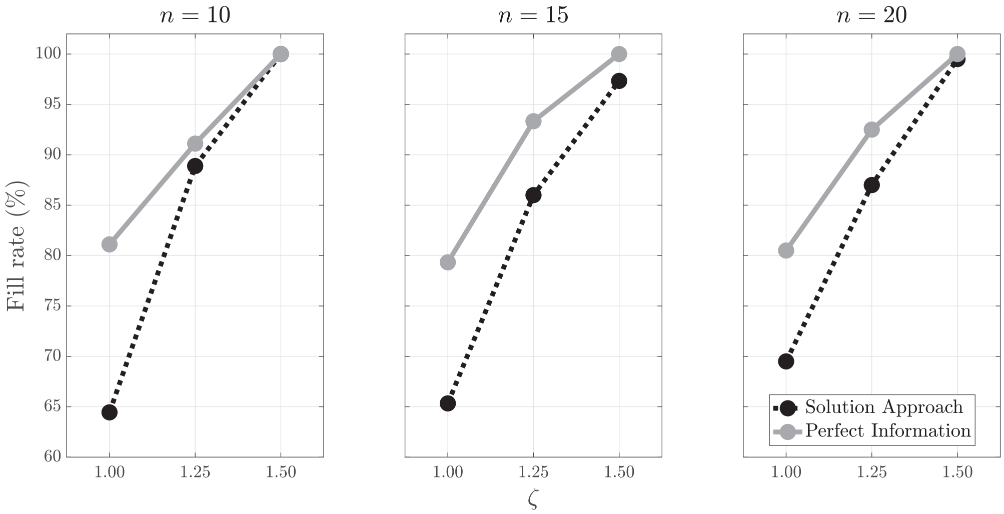


Fig. 6. Average fill rates (%) for different instance sizes (n) and values of ζ, considering DOD = 75%.

greedy policy exhibits a higher acceptance rate at the beginning of the day but decreases rapidly before reaching the middle of the delivery horizon. The benefits of rejecting early orders to increase the number of future orders accepted can be clearly seen in the clustered customer distributions C101 and C201, which allow an average fill rate increase of about 8.8% and 6.2%, respectively (see Table 8).

Table 8
Average fill rates over 10 instances for different customer distributions, considering 100 customers, DOD = 75%, and $\zeta = 1$.

Distribution	Policy	
	π^{A}	π^{G}
R101	68.1	65.7
C101	62.5	53.7
C201	61.9	55.7
RC101	59.3	55.0

Analysis of drone resupply. We now turn our attention to understanding the value of drone resupply over a more conventional resupply approach. We consider a case where the delivery truck is resupplied by another truck with the same operational characteristics (i.e., an uncapacitated truck with the same travel speed). Intuitively, this truck-based resupply strategy is faster than the case where the delivery truck must return to the depot to collect newly arrived orders (the approach commonly studied in the literature; see Section 2). This is because delivery and resupply activities can occur in parallel, allowing the delivery truck to continue serving customers while the other truck handles the resupply of new orders. Thus, the results presented below represent a best-case scenario for truck-based resupply in terms of orders covered using a single resupply vehicle, which we compare with the proposed drone resupply.

Fig. 8 shows the relative improvements that the drone-based resupply strategy achieves over the conventional, truck-based resupply system (both systems are controlled by the adaptive policy). Results show that drone resupply increases fill rates in 39 out of 40 tested instances (in one C101 instance, due to the unlimited capacity of the resupply truck, the truck-based resupply system delivers one more order than the drone resupply system). The maximum relative improvement reached by drone resupply is 21.1%, equivalent to 12 more customers served. Comparatively, drone resupply is less advantageous for the RC101 customer distribution (e.g., the median relative improvement is 3.6%, compared to 7.9% achieved for the R101 customer distribution), which is consistent with the fill rates shown in Table 8. Nevertheless, note that the benefits of using aerial drones instead of conventional ground vehicles for resupplying extend even further when considering potential positive effects for urban logistics, such as reducing traffic congestion and emissions.

To further understand the behavior of drone resupply (under the adaptive policy), Table 9 reports average metrics for different values of ζ (the factor that defines the length of the delivery horizon). For the R101 distribution, fill rates exhibit roughly the same behavior as those shown in Fig. 6. In C101 instances, the drones make fewer sorties but their capacity utilization (in terms of orders resupplied per sortie) is higher. Similar to Table 8, fill rates decrease about 4%–6% when customers are clustered. The inefficiencies that arise when serving customers with a non-uniform spatial distribution can also be observed in rows "Idle time (%)" (i.e., the percentage of the total distribution time that the truck is idle), which increases in the C101 instances. Such distributions strongly benefit by having a longer delivery horizon, as shown in Table 9 and Fig. 6.

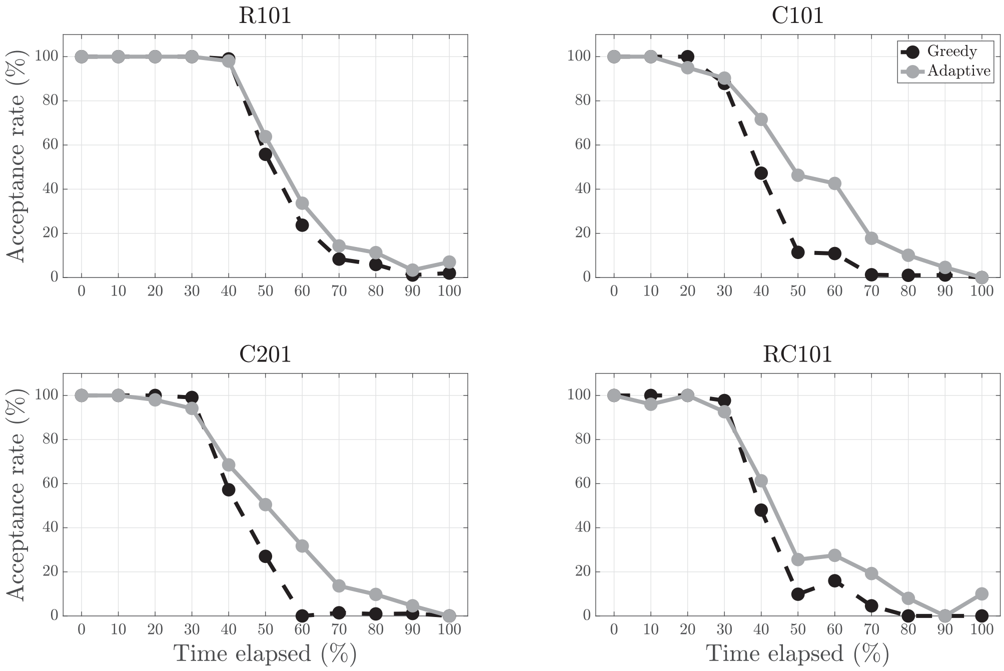


Fig. 7. Average acceptance rate (%) over the percentage of the delivery horizon that has elapsed, considering 10 instances of 100 customers, DOD = 75%, $\zeta = 1$, and two drones with a load capacity of four orders each.

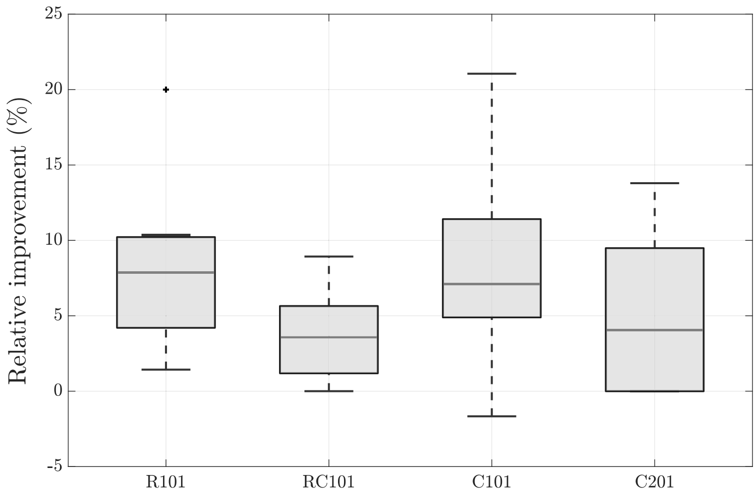


Fig. 8. Distribution of the relative improvement (%) achieved by the drone-based resupply system (with two drones with a load capacity of four orders each) over the uncapacitated truck-based resupply system, considering DOD = 75% and $\zeta = 1$.

Decision times. Finally, we report the computational time needed to make each decision, which is defined as the sum of the time to insert the newly arrived order (through the Cheapest Insertion technique) and the time to solve Problem (DRP) to optimality. Considering only those decision epochs when Problem (DRP) was invoked (see Section 5.2), the average decision time (over the four Solomon's customer distributions) was 0.022 s, while the maximum decision time was 0.13 s, reached when making a single decision in one C101 instance with $\zeta = 1.50$ (see our online repository for further details). These results clearly show that our approach respects the tight time frames imposed by real-time decision-making, thus allowing same-day delivery providers to provide immediate feedback to customer requests. As discussed in Section 2, only five works on DVRPs consider real-time decision-making, with only two reporting computational times to make each decision (and these exceed several seconds). Moreover, works addressing the truck-only scenario in which trucks resupply themselves by returning to the depot, such as Klapp et al. (2018) and Ulmer et al. (2019), make decisions every time the truck returns to the depot or stops at a customer location, thus delaying the acceptance/rejection of new orders until such events occur, which is clearly operationally untenable.

Table 9
Average metrics of the adaptive policy over 10 instances of 100 customers, considering DOD = 75% and two drones with a load capacity of four orders.

Distribution	Metric	ζ		
		1.00	1.25	1.50
R101	Fill rate (%)	68.1	82.8	97.2
	Total drone sorties	14.5	17.1	20.0
	Orders shipped per sortie	3.0	3.4	3.6
	Idle time (%)	2.2	1.9	2.5
C101	Fill rate (%)	62.5	76.7	93.0
	Total drone sorties	12.1	14.5	17.5
	Orders shipped per sortie	3.1	3.6	3.9
	Idle time (%)	3.5	4.6	4.6

7. Conclusions

We introduced the Dynamic Traveling Salesman Problem with Drone Resupply (DTSP-DR), which consists of dynamically routing a single truck that can receive newly arrived orders along its route via drones dispatched from the depot. This problem arises in same-day delivery services, where providers must plan their distribution while simultaneously receiving new orders. A particularly challenging aspect is adhering to the stringent time frames of real-time decision-making, as customers expect immediate feedback on their requests.

We formulated the problem as an MDP, where decisions are made each time a new order arrives. Then, we proposed an online policy based on reoptimization, which combines an insertion heuristic with a fast MILP model to adjust drone resupply operations optimally. To explore the performance of the solution approach, we developed a MILP formulation for the static counterpart of the DTSP-DR, where customer information is completely known before the operation starts (i.e., the perfect-information scenario).

Numerical experiments showed that our solution approach has a maximum average fill rate decrease of 20% over the perfect-information counterpart. For large instances, we show that our online policy has a fill rate increase of up to 8% over a greedy policy (where any order that can be feasibly served is accepted). We also show that the drone-based resupply strategy increases fill rates by up to 21% compared to a more traditional system where the delivery truck is resupplied by another truck with the same operational characteristics. For instances of 100 customers, computational times to make each decision are in the hundredths of a second, which is appealing considering the use of an exact approach to reoptimize drone resupply operations. This allows real-time feedback to customers regarding their eligibility for SDD. In summary, we defined a practical problem in last-mile drone logistics, devised an effective and computationally expedient solution methodology, and computationally demonstrated its behavior on established instances from the literature.

Future work includes extending the MDP and the proposed solution approach to allow the truck to return to the depot multiple times during the day to collect newly arrived orders. This would also enable us to make a direct comparison with the scenario where trucks resupply themselves by returning to the depot, as current related works do not consider real-time decision-making (see, e.g., Klapp et al. (2018) and Ulmer et al. (2019)). Furthermore, the DTSP-DR aims to maximize the total number of SDD orders accepted. This implicitly assumes that rejected orders are offered delivery on a later day, without considering the potential impact of those decisions on the following day's operations. Therefore, future work might extend the DTSP-DR to consider a multi-day planning horizon. To our knowledge, only Banerjee et al. (2023) (in a working paper) have addressed a similar problem but without considering real-time decision-making or drone resupply. Another extension is considering a specific profit (or penalty) for accepting (or rejecting) a given customer request, which might depend on the location or other characteristics of the customer. Finally, the DTSP-DR can be extended by considering multiple delivery trucks serviced by a fleet of drones.

CRediT authorship contribution statement

Juan C. Pina-Pardo: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Daniel F. Silva:** Conceptualization, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **Alice E. Smith:** Conceptualization, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing. **Ricardo A. Gatica:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data has been shared on GitHub https://github.com/jcpinap/DTSP-DR.

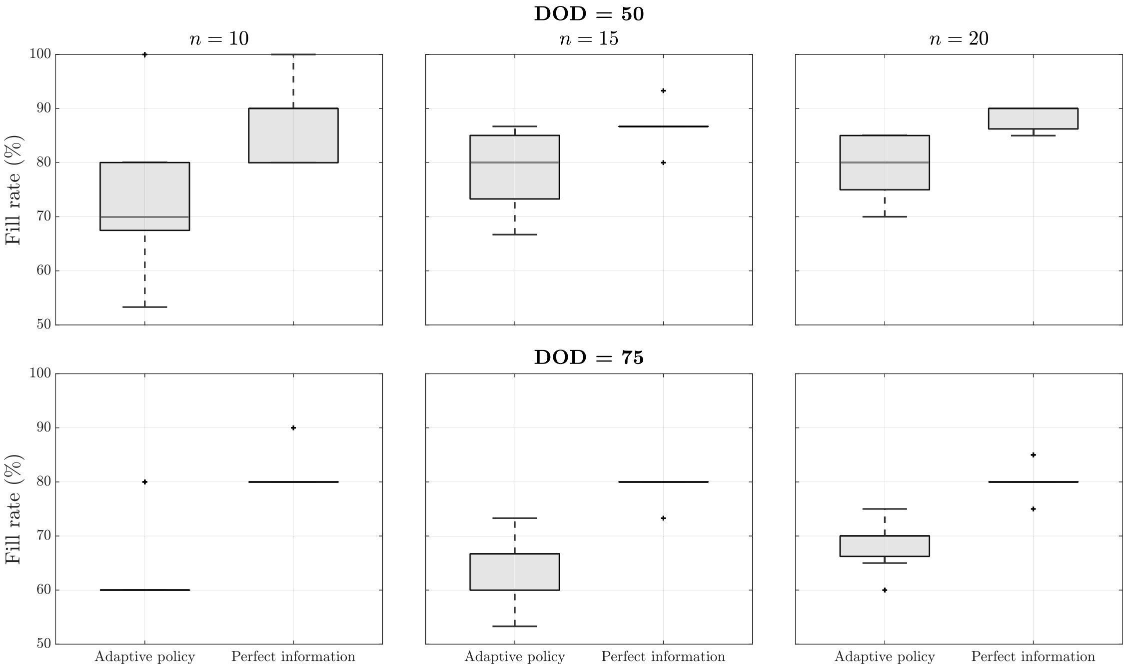


Fig. A.9. Distribution of fill rates (%) for different instance sizes (n) and degrees of dynamism, considering the R101 customer distribution, $\zeta = 1$, and two drones with a load capacity of four orders each.

Appendix. Supplementary material

Effectiveness of valid inequalities. Considering the R101 customer distribution, $\zeta = 1$, DOD = 50, two drones with a load capacity of four orders each, and a maximum run-time of 600 s, Table A.10 shows the effectiveness of valid inequalities (42)–(47) in reducing both run-times and optimality gaps. For instances of 15 customers, the MILP model without the valid inequalities was not able to find any optimal solution in 600 s.

Table A.10
Results of the MILP model (14)–(47) with and without valid inequalities (42)–(47).

Customers	Instance	Without valid inequalities		With valid inequalities	
		Run-time (s)	Gap (%)	Run-time (s)	Gap (%)
10	1	387.69	0.0	0.16	0.0
	2	309.19	0.0	0.11	0.0
	3	22.31	0.0	0.05	0.0
	4	395.73	0.0	0.20	0.0
	5	242.91	0.0	0.11	0.0
	6	375.06	0.0	0.27	0.0
	7	391.27	0.0	0.19	0.0
	8	305.84	0.0	0.09	0.0
	9	48.30	0.0	0.09	0.0
	10	304.53	0.0	0.13	0.0
	Average	**278.28**	**0.0**	**0.14**	**0.0**
15	1	600.09	36.4	1.77	0.0
	2	600.13	36.4	1.11	0.0
	3	600.13	36.4	0.84	0.0
	4	600.08	36.4	0.89	0.0
	5	600.11	25.0	3.31	0.0
	6	600.14	25.0	6.45	0.0
	7	600.13	25.0	1.38	0.0
	8	600.09	36.4	1.14	0.0
	9	600.09	15.4	1.61	0.0
	10	600.09	25.0	0.89	0.0
	Average	**600.11**	**29.7**	**1.94**	**0.0**

Table A.11
Notation for the MDP formulation.

Symbol	Description
a_τ	Action at decision epoch τ. Formally, $a_\tau := (z^\tau, \mathrm{T}^{\tau,a}, \mathrm{D}^{\tau,a})$.
d_n	Drone flying time between the depot and node $n \in \mathcal{N}^{\tau,a}$.
$m^\tau_{k(i)}$	Meeting location (with the truck) associated with the ith sortie of drone $k \in K$.
$r(s_\tau, a_\tau)$	Reward of taking action a_τ at system state s_τ.
s_τ	System state at decision epoch τ. Formally, $s_\tau = (\lambda^\tau, \mathcal{A}^\tau, \mathrm{T}^\tau, \mathrm{D}^\tau)$.
$s_{\tau,a}$	Post-decision state at decision epoch τ under action a_τ.
$t_{n,n'}$	Truck travel time between nodes n and n', with $n, n' \in \mathcal{N}^{\tau,a}$.
z^τ	Order acceptance indicator (i.e., $z^\tau = 1$, if the order is accepted; $z^\tau = 0$, otherwise).
α^τ_k	Time when drone $k \in K$ will be available at the depot to start a new sortie.
δ	Time the truck needs to receive a drone en route.
ϑ^τ	Order received at decision epoch τ.
λ^τ	Time at which decision epoch τ occurs.
μ_τ	Function that maps each state s_τ into an action $\mu_\tau(s_\tau) = a_\tau$.
$\xi^\tau_{k(i)}$	Scheduled launch time of the ith sortie of drone $k \in K$.
τ	Decision epoch counter.
$\mathcal{A}^\tau$	Set of accepted orders that are available at the depot at decision epoch τ.
D^τ	Drone vector information at decision epoch τ. Formally, $\mathrm{D}^\tau = (\mathrm{D}^\tau_k)_{k\in K}$.
D^τ_k	Vector information of drone $k \in K$ at decision epoch τ. Formally, $\mathrm{D}^\tau_k := (\alpha^\tau_k, \mathcal{O}^\tau_k, \mathrm{P}^\tau_k)$.
$\mathcal{F}^\tau$	Set of orders delivered between decision epochs τ and $\tau + 1$.
H	Delivery horizon (i.e., the time when the truck must return to the depot).
$\mathcal{I}^\tau_k$	Set of sortie indices associated with drone $k \in K$ at decision epoch τ.
$\mathbb{I}(n)$	Indicator function that takes the value 1 if a drone that will resupply the truck at node $n \in \mathcal{N}^{\tau,a}$, and 0 otherwise (i.e., $\mathbb{I}(n) = 1$ if and only if $n \in \{m^{\tau,a}_{k(i)}\}_{k\in K, i\in \mathcal{I}^{\tau,a}_k}$).
K	Set of resupply drones.
$\mathcal{M}^{\tau,a}_{k(i)}$	Set of all possible meeting locations where drone $k \in K$ can send orders in $\mathcal{R}^{\tau,a}_{k(i)}$ according to the drone flight endurance.
$\mathcal{N}^\tau$	Set of nodes visited in the current truck route. Formally, $\mathcal{N}^\tau := \mathcal{L}^\tau \cup \mathcal{O}^\tau \cup \mathcal{A}^\tau \cup \{0\}$.
$\mathcal{L}^\tau$	Set of orders currently loaded onto the truck at decision epoch τ.
$\mathcal{O}^\tau$	Set of orders currently onboard all drones at decision epoch τ.
$\mathcal{O}^\tau_k$	Set of orders onboard drone $k \in K$ at decision epoch τ.
P^τ_k	Vector of planned sorties for drone $k \in K$ at decision epoch τ. Formally, $\mathrm{P}^\tau_k := \left(\mathrm{P}^\tau_{k(i)}\right)_{i\in \mathcal{I}^\tau_k}$.
$\mathrm{P}^\tau_{k(i)}$	The ith planned sortie. Formally, $\mathrm{P}^\tau_{k(i)} := (\mathcal{R}^\tau_{k(i)}, m^\tau_{k(i)}, \xi^\tau_{k(i)})$.
$\mathcal{R}^\tau_{k(i)}$	Subset of orders to be resupplied in the ith sortie of drone $k \in K$.
$\mathcal{S}^\tau$	Set of orders resupplied between decision epochs τ and $\tau + 1$.
T_n	Truck departure time from node $n \in \mathcal{N}^{\tau,a}$.
T^τ	Truck vector information at decision epoch τ. Formally, $\mathrm{T}^\tau := (\mathcal{X}^\tau, \mathcal{L}^\tau)$.
$\mathcal{T}$	Random variable representing the number of decision epochs.
$\mathcal{X}^\tau$	Truck route at decision epoch τ visiting nodes in $\mathcal{N}^\tau$.

Detailed results on fill rates. Fig. A.9 shows the distribution of the fill rates for the adaptive policy and the perfect-information model. Overall, the fill rates achieved by the adaptive policy approach get closer to the perfect information scenario as the number of customers requesting same-day delivery service increases.

Notation for the MDP formulation. Table A.11 shows the notation used throughout Section 3.

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